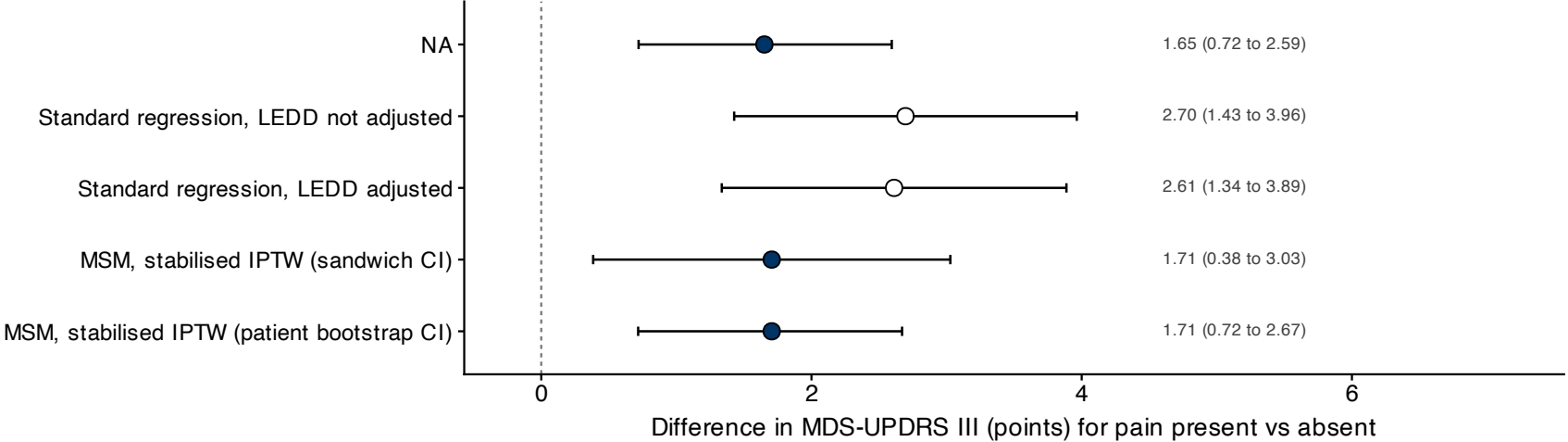


Figure 4. Handling dopaminergic dose as a time-varying confounder



Dopaminergic dose is affected by prior pain and in turn affects both later pain and motor severity, so conventional regression is biased whether or not it is adjusted for. Filled circles are the two causal estimators, which agree closely with each other and are about 37% smaller than the conventional estimates. All five models are fitted on the same 1,970 observations from 787 patients. LEDD, levodopa equivalent daily dose.